

BLAZE A TRAIL 3.0

- **Problem Statement Title** - AI-Enabled Smart Search-and-Rescue Ground Robot
- **PS Category**- Hardware
- **Team Name** - IntelliX Titans



“Saving Lives Through Intelligent Robotics”



We Make You Shine
St. JOSEPH'S INSTITUTE OF TECHNOLOGY
(An Autonomous Institution)
St. Joseph's Group of Institutions
OMR, CHENNAI - 119



RESQBOT (S.A.R.A)

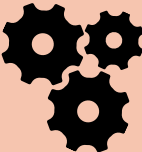
PROPOSED SOLUTION

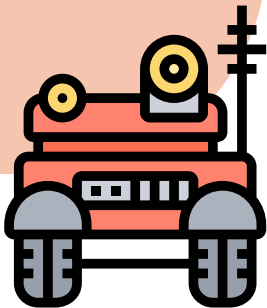
We propose a **low-cost AI-assisted** Search and Rescue Robot capable of **navigating** disaster environments, **detecting** survivors, **identifying hazardous** gas leaks, transmitting **GPS coordinates**, and providing **live video** to rescue teams.

INNOVATION

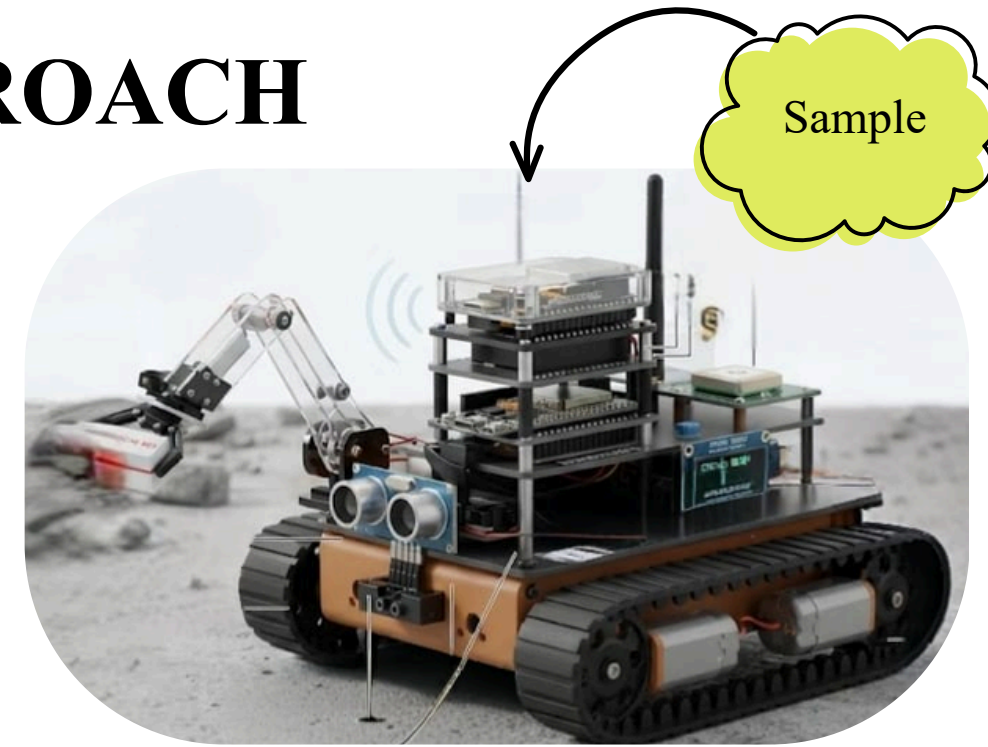
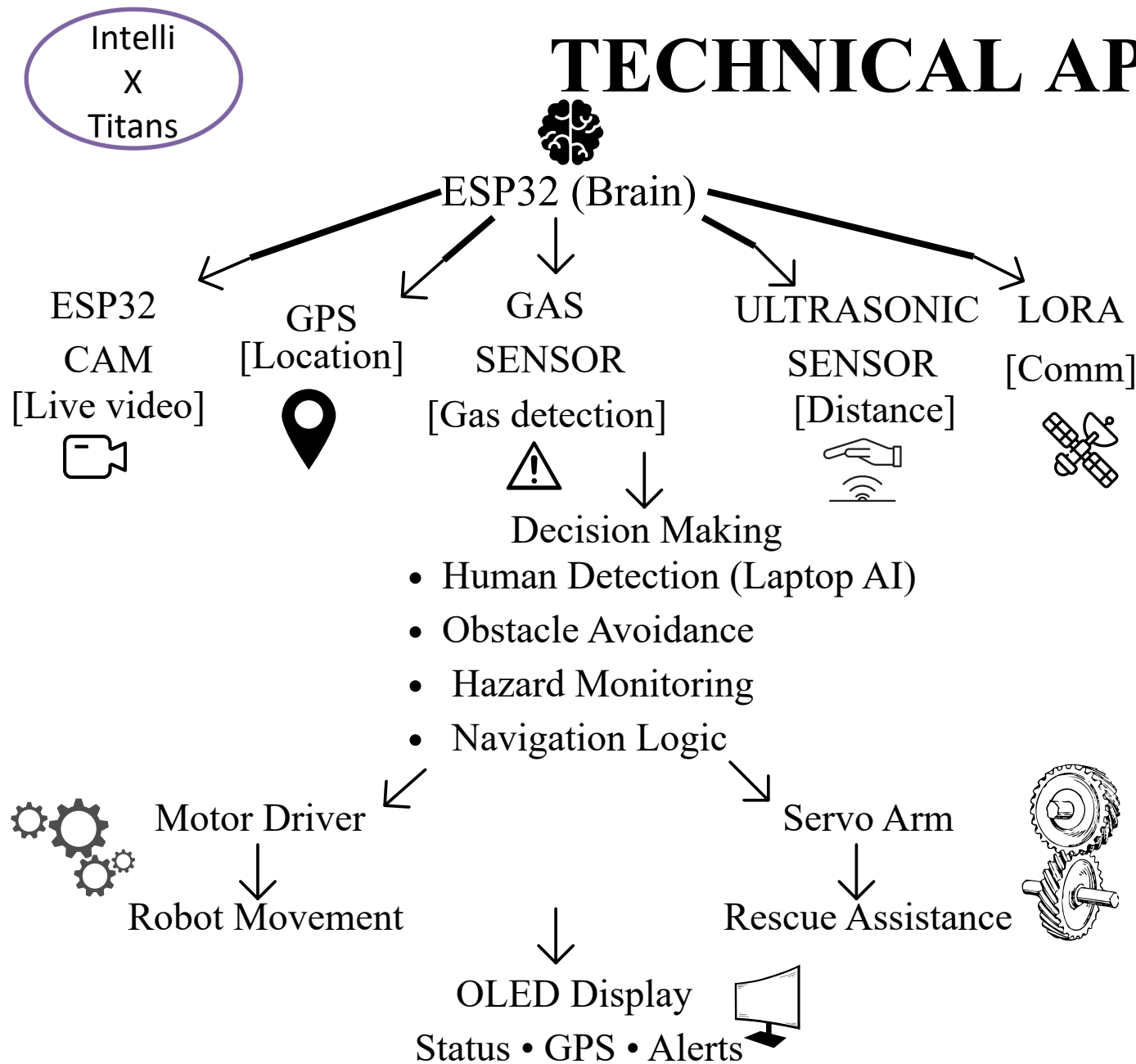
A modular **rescue** robot built using **affordable hardware**, making it **accessible** for **educational institutions** and **emergency response** research.

KEY FEATURE'S

- **Autonomous** obstacle avoidance
- GPS-based location **tracking**
- ESP32-CAM **live video** streaming
- Hazardous gas **detection**
- LoRa-based **long-range communication** 
- Servo-operated **rescue** mechanism
- OLED status **display**



TECHNICAL APPROACH



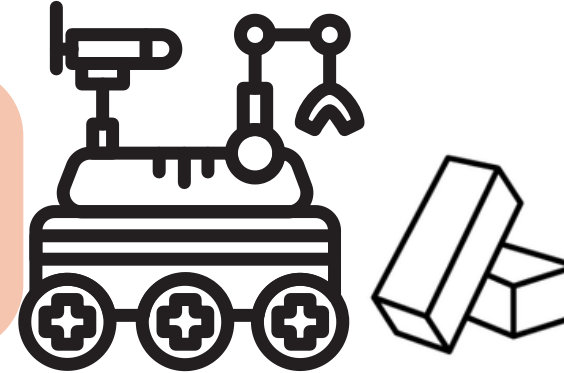
Development Flow

Arduino IDE → Program ESP32
 Python ← VS Code ← Robot Movement
 ↓
 OpenCV + YOLO → Human Detection

FEASIBILITY AND VIABILITY

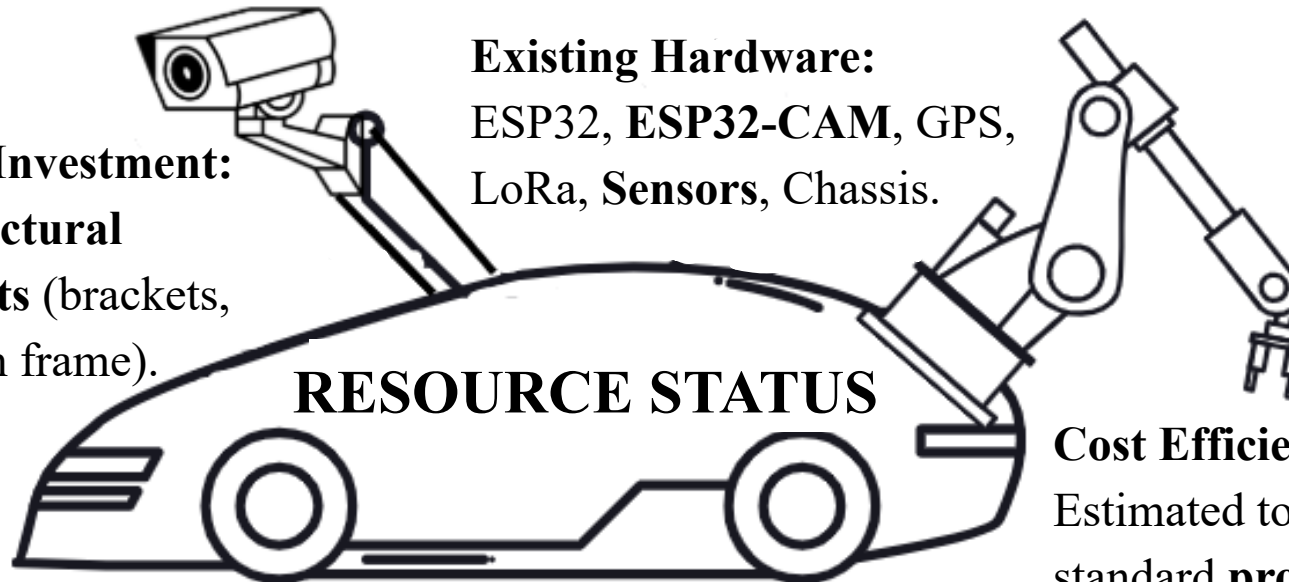
Operational Resilience:

- Auto-returns to **base** via last-known GPS if LoRa signal is lost.
- Offloads heavy processing to a **laptop** for agility and efficiency.
- Tracked chassis **ensures** superior traction on **uneven disaster** terrain.



Required Investment:
Minor **structural components** (brackets, robotic arm frame).

Existing Hardware:
ESP32, **ESP32-CAM**, GPS, LoRa, **Sensors**, Chassis.

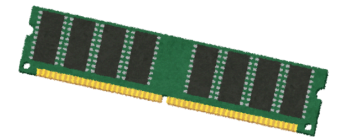
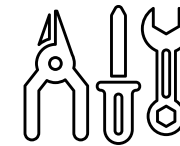


RESOURCE STATUS

Cost Efficiency:

Estimated total **build cost** is roughly **30%** of standard **professional-grade** research platforms.

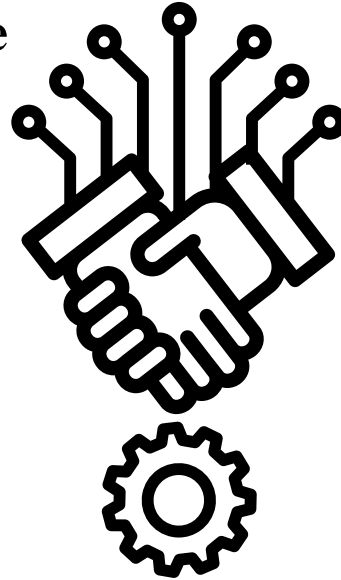
DEVELOPMENT STRATEGY:
Modular **subsystem** integration
(Test one **sensor** at a time).



IMPACT AND BENEFITS

IMPACTS

- AI robot improves **emergency rescue**
- *Reduces rescue time* significantly
- Keeps rescue *teams safer*
- *Finds survivors* more quickly
- *Tracks location* and surroundings
- *Supports faster* rescue decisions



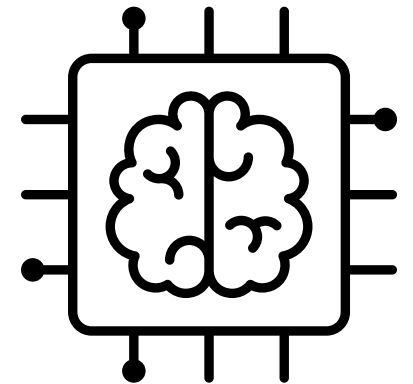
FUTURE SCOPES

- *Thermal Camera* for night-time survivor detection
- *Satellite Communication* for remote disaster zones
- Multi-DOF *Robotic Arm* for debris removal
- *AI-based Mapping* and Autonomous Navigation
- Cloud Dashboard for *real-time monitoring*
- Drone Integration for *aerial support*

BENIFITS

- *Faster* Victim Detection
- *Accurate Location* Tracking
- *Hazard Detection*

- Modular & *Scalable Design*
- Reliable *Long-Range* Communication
- *Low-Cost* & Affordable



RESEARCH AND REFERENCES

1. Casper, J., & Murphy, R. R. (2003).

Research Focus: Human–robot collaboration, victim search, disaster response robotics.

IEEE Xplore: <https://ieeexplore.ieee.org/document/1197955>

2. Murphy, R. R. (2012).

Research Focus: Evolution of rescue robots, autonomous systems, disaster deployments.

IEEE Xplore: <https://ieeexplore.ieee.org>

3. **Authors:** Robin R. Murphy, Satoshi Tadokoro

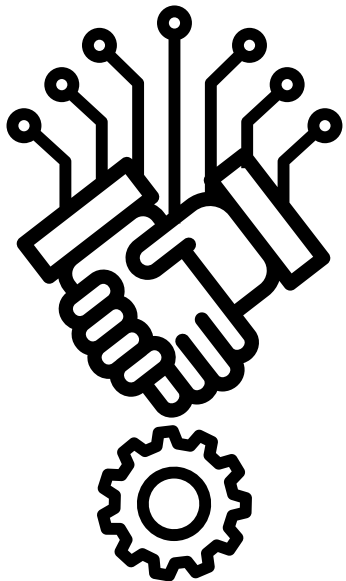
Research Focus: Search-and-rescue robot design, sensing, localization, communication, and rescue applications.

DOI: https://doi.org/10.1007/978-3-540-30301-5_51

4. **Purpose:** Literature review and analysis of recent research papers on AI, computer vision, disaster robotics, GPS navigation, and autonomous mobile robots.

Link: <https://scholar.google.com>

TEAM MEMBERS DETAILS



Jeffisha A	'III' Year	ECE
John Stewartsson J R	'III' Year	ECE
Joshua Miracle J	'III' Year	ECE
Mahalakshmi P	'III' Year	ECE
Namith S P	'III' Year	ECE
Narmathakavi C S	'III' Year	ECE